

Synopsis of the B.E. Dissertation

VentureIQ: An Agentic AI and RAG-Powered Intelligent Startup Investment Platform

to be submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering

in

Computer Science and Business Systems

by

Devansh Sangwan	1BM23CB014
Paras	1BM23CB032
Vedit Agrawal	1BM23CB060
Dimple Hirwani	1BM23CB065

Name of the Internal Guide

Dr. Gururaja H.S.

Department of Computer Science and Business Systems



BMS College of Engineering
Bull Temple Road, Basavanagudi.
Bangalore 560019

June, 2026



BMS COLLEGE OF ENGINEERING

**(Autonomous College under VTU)
Bull Temple Road, Basavanagudi,
Bangalore –560019**

Department of Computer Science and Business Systems

C E R T I F I C A T E

This is to certify that the **Project Phase I** report entitled **VentureIQ: An Agentic AI and RAG-Powered Intelligent Startup Investment Platform** is a bona-fide work carried out by **Devansh Sangwan (1BM23CB014), Paras (1BM23CB032), Vedit Agrawal (1BM23CB060) and Dimple Hirwani (1BM23CB065)** in partial fulfilment for the award of degree of Bachelor of Engineering in **Computer Science and Business Systems** from **Visvesvaraya Technological University, Belgaum** during the year **2025-2026**. The report has been approved as it satisfies the academic requirements in respect of technical activity prescribed for the Bachelor of Engineering Degree.

Marks Obtained	
Maximum Marks	

Signature of the Guide
Dr. Gururaja H.S.
Guide
Dept of CSBS, BMSCE

Signature of the HoD
Dr. R. Ashok Kumar
Professor and HoD
Dept of CSBS, BMSCE

Table of Contents

1. Abstract.....	2
2. Introduction.....	3
3. Review of Literature	4
3.1 Review of Existing Platforms	4
3.1.1 AngelList (Wellfound).....	4
3.1.2 Crunchbase.....	5
3.1.3 PitchBook.....	5
3.1.4 Gust	6
3.1.5 Carta	7
3.2 Related Research Papers.....	9
3.2.1 A Fused Large Language Model for Predicting Startup Success.....	9
3.2.2 Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG.....	10
3.2.3 FinBERT: Financial Sentiment Analysis with Pre-trained Language Models	11
3.2.4 Predicting Startup-VC Fund Matches.....	12
4. Problem Definition.....	14
5. Objectives of the Project Work.....	15
6. Requirement Specification	17
6.1 Functional Requirements	17
6.2 Non-Functional Requirements.....	18
6.3 Technical Stack.....	18
7. System Design	19
8. Methodologies.....	22
8.1 Startup Data Pipeline	23
8.2 VC/Investor Data Pipeline	24
8.3 AI Output and Scoring System	25
9. Conclusion	27
10. References.....	28

1. Abstract

India has experienced rapid growth in its startup ecosystem during the last ten years because of digital advancements and government programs and increased investor activity. The startup investment process continues to experience significant problems because of multiple operational platforms which depend on unverified self-reported information and need extensive manual verification. Existing platforms such as AngelList, Crunchbase, and PitchBook mainly focus on fundraising, networking, or data aggregation, but do not provide a unified system for startup verification, intelligent evaluation, and investor decision support.

VentureIQ - An Agentic AI and RAG-Powered Intelligent Startup Investment Platform operates as a technology-driven startup investment intelligence platform which helps investors assess startups through its improved evaluation process that delivers greater transparency and data-based results. It bridges this critical gap by combining Agentic AI, Retrieval-Augmented Generation (RAG), multi-dimensional machine learning scoring, and a compliant investment marketplace into a single unified platform.

It brings together multi-source data verification, AI-assisted analysis, fraud detection, and personalized investor-startup matching within a single integrated system. The system consists of two interrelated pipelines which include the Startup Data Pipeline for processing startup registrations and financial documents and external data sources and the VC/Investor Data Pipeline for studying investor preferences together with their actual market activities.

The system uses document intelligence models, external verification sources, and machine learning-based anomaly detection to identify inconsistencies, evaluate startup credibility, and generate explainable investment insights across factors such as growth potential, risk level, and founder credibility. VentureIQ also proposes a compliant marketplace layer for secure transactions with verification and escrow-based mechanisms.

By integrating verification, intelligent analytics, and explainable AI into one platform, VentureIQ aims to reduce information asymmetry and support more efficient, transparent, and trustworthy startup investment decision-making.

2. Introduction

The current funding of early-phase Indian startups requires investors to verify information through manual methods because they have no trust-based systems which prevent information fragmentation and trust-based systems. Investors use pitch decks and LinkedIn profiles and company records and financial statements to assess startups but they lack tools which can confirm the accuracy of these sources. The absence of complete data together with personal judgment leads to incorrect investment decisions. The existing platforms - AngelList and Crunchbase and PitchBook do not offer a complete solution for the investment process because they only assist with fundraising and networking and data collection activities.

The startup investment intelligence platform VentureIQ uses Agentic AI together with RAG technology to create a system that evaluates startups through reliable transparent data-driven methods. A document intelligence system processes startup registration documents and pitch decks and financial documents while an autonomous agentic enrichment layer conducts cross-validation using sources which include MCA21 and GST filings and LinkedIn and GitHub and WHOIS records. The platform unites multiple functions which include multi-source verification and AI-driven analysis and fraud detection and investor-startup matching into one complete system. It also uses anomaly detection models to examine startup profiles which investors will access to find reporting patterns that appear suspicious and show inconsistencies.

VentureIQ integrates investor analysis which combines investment preference data and behavioral activity data to give startup recommendation and discovery results. The platform utilizes a Retrieval-Augmented Generation (RAG) framework which creates contextual insights through startup comparisons with historical and market data. In addition the platform proposes a marketplace system which meets legal requirements will enable Know Your Customer (KYC) checks, use escrow funds for transactions, and implement Right of First Refusal (RoFR) systems while following Indian regulatory requirements. VentureIQ built its platform with verification systems and explainable AI and intelligent recommendation capabilities to create a system which enables investors to make better investment decisions through transparent verification processes. These methods ensure a complete background verification of the startup, its numbers, and its history making our platform an end to end ideal deal for reliable and trustworthy engagements.

3. Review of Literature

3.1 Review of Existing Platforms

A structured review of major existing platforms reveals the fragmented state of the startup investment ecosystem. Each platform addresses one or two aspects of the investment lifecycle but none integrates data verification, AI intelligence, and investment execution into a unified system.

3.1.1 AngelList (Wellfound)

AngelList, which runs under the name **Wellfound** for its **recruitment purposes**, is an **online portal** that helps **entrepreneurs raise capital** and **recruit talent**, while at the same time offering opportunities for **investors** to get involved. The platform offers three main functionalities: **startup financing**, **syndicate investment**, and **startup recruiting** – making it possible for **entrepreneurs and investors** to meet and interact directly **without any intermediaries**.

Key Features:

- **Direct Access to Investors:** Helps entrepreneurs access funding from investors through syndicate investments and rolling funds without any intermediaries.
- **Hiring Marketplaces:** Through its subsidiary Wellfound, startups can quickly post openings and recruit talent without having to rely on headhunters.
- **Tools for Investor Discovery:** Enables investors and entrepreneurs to discover each other through profiles and start deals right away without any warm introductions.
- **SPV, Rolling Funds & Venture Funds:** Helps entrepreneurs set up SPVs, rolling funds, and venture funds, including registration, filings, Know Your Customer (KYC) and Anti-Money Laundering (AML) compliance, tax forms, and LP payments.

Major Drawbacks:

- **Voluntary Startup Information:** Almost all information about startups comes voluntarily from the founders, without any validation, increasing the chances of overstated figures.
- **No Automated Score:** The platform lacks an automated scoring model to evaluate the worthiness of a startup or provide rankings for investors' consideration.
- **Incomplete Fraud Prevention:** Although AngelList conducts KYC and AML procedures on LPs, there is no verification mechanism of startup claims to ensure accuracy.
- **Connectivity Bias:** Successful startups that are known to the public have more visibility within the network than new or less-known ones.

How It Helped Shape VentureIQ:

AngelList showed us how a single platform enables efficient startup and investor connections. The platform serves as an investment solution while it checks the trustworthiness of startup claims. This became one of the main inspirations behind VentureIQ, where we focused on verification, fraud detection, explainable AI scoring, and more trustworthy startup evaluation.

3.1.2 Crunchbase

Crunchbase is a comprehensive business intelligence platform used by investors, sales professionals, entrepreneurs, and researchers to access data on companies, funding rounds, investors, and industry trends. It serves as a primary tool for market research, competitive analysis, lead generation, and venture capital deal sourcing, helping users discover private market opportunities and track startup activity.

On Crunchbase startups can present their venture to investors through a detailed online profile. The online profiles can include both (i) fundamental variables, which provide structured information on founders, funding, and the business sector, and (ii) textual self-description.

Key Features:

- **Helps in:** Discovering Startups, Provides Company's Intelligence, Tracking Funding & Investments, Investor Discovery.
- **Deep-Dive Search:** Filters by niche, city, funding round etc and pinpoint your next move.
- **Trend Analysing:** Reveals where big bets are landing, like AI or climate tech surges.
- **Detailed Company Profiles:** Access to extensive data including funding history, leadership teams, product offerings, acquisitions, and financial metrics for startups and established corporations.
- **AI-Powered Insights:** Crunchbase Pro includes AI Search using natural language processing, Crunchbase Scout for quick research summaries, and personalized homepage feeds with suggested actions.

Major Drawbacks:

- **Data Quality Issues:** Users report inconsistent data, outdated entries, and high rates of invalid contact emails (up to 40% in some cases), reducing reliability for direct outreach.
- **Limited International Coverage:** The database is heavily US-focused, with sparse or inaccurate firmographic data for businesses in regions like the UK, India, etc.
- **Free Tier Limitations:** The free version offers limited functionality, while paid plans can be costly for smaller users with less intuitive export options compared to competitors.
- **Zero Brainpower:** Database, not advisor i.e. no scam flags or predictive scores.
- Crunchbase does not have investment capability i.e. an investor cannot invest in a startup on their platform.

How It Shaped VentureIQ:

Crunchbase showed data's only as good as its truthfulness, but the blind-trust vibe is never good. It inspired us to build VentureIQ with source-crossing verifiers and fraud radars, turning raw lists into decisions you can bank on.

3.1.3 PitchBook

A leading provider of financial research and data analytics for venture capital firms, private equity firms, and institutional investors, PitchBook offers information about funding activities, valuation, mergers & acquisitions, as well as investments.

Key Features:

- **The Hard Numbers:** Access to detailed data on firms, funds, deals, and M&A, including privately-held valuations.
- **Visual Analytics:** Clean visuals with charts & tables organized by geographic location, type of fund, and market dynamics.
- **Company and Firm Digs:** A firm-level view of all the past and current activities, fund performance, and portfolio company data.
- **Live Portfolio Tracking:** An overview of portfolio company performance & deal flow, available via customisable dashboard.
- **AI Scoring Tools:** VC Exit Predictor ranks startups based on their likelihood of going public & being acquired, while PitchBook Navigator gives answers to any market-related questions with the help of AI technology.

Major Drawbacks:

- **Expensive Pricing Structure:** Designed for large organizations; small investors, students, and start-up founders cannot afford it.
- **Incomplete Data:** Many profiles lack important information, especially those of small and/or young firms, and nothing can be validated.
- **Score Calculation Issues:** The artificial intelligence algorithm suffers from certain biases and struggles to cope with unexpected circumstances such as global market turmoil and catastrophes.
- **Lack of Transactions:** This database is purely academic and serves only for research purposes, and not for trading and investing.
- **No Fraud Prevention and No Investor Matching:** The tool offers access to relevant data but provides no additional protection layer and offers no means to find investors.

How It Shaped VentureIQ:

PitchBook showed us that their statistical data is excellent. The way they present data from "big firms" excludes other players from competition. We altered our initial design for VentureIQ to create a better user experience. We focused on a verification process and insights from artificial intelligence, which PitchBook does not offer to support regular investors.

3.1.4 Gust

Gust is a startup investment platform which mainly supports early-stage startups and angel investors and startup accelerators. The platform enables startups to develop organized profiles which they use to seek funding through online investment networks. Angel groups and startup incubators use the platform to handle their application processes and track their deal flow activities.

Key Features:

- **Startup Profile Creation:** It allows startups to build structured investment profiles.
- **Angel Network Access:** Helps startups connect with angel investors and seed funds.
- **Application Management:** Supports startup applications for accelerators and investment programs.

- **Deal Flow Management:** Helps investor groups organize and review startup opportunities.

Key Limitations:

- **Limited Verification:** Startup information is self-reported.
- **Weak Fraud Detection:** Has no cross-checking of startup claims.
- **Heavy Human Dependency:** Investment decisions depend mainly on investor judgment.

How It Helped Shape VentureIQ:

Gust helped us see how important structured onboarding for startups and organized deal flow management are. However, we also noticed that the platform relies a lot on manual investor evaluation without smart analysis or checks. This led VentureIQ to focus on automated verification, fraud detection, and clear AI-supported decision-making.

3.1.5 Carta

Carta serves as an investment platform for startups which provides its main services to early-stage startups and angel investors and startup accelerators. The system enables startups to develop organized profiles which they can use to request funding from online investment platforms. The platform serves its primary function for angel groups and startup incubators who use it to handle their application processes and manage their deal flow operations.

Key Features:

- **Cap Table Sanity:** It keeps a live look at shares and dilution so founders actually know who owns what at any given second.
- **Automation:** Carta has automated the valuation reports and financials you need to stay compliant.
- **Fund Administration:** makes life way easier for venture LPs and GPs by letting GPs manage the entire fund in one platform.
- **Secondary Market:** Through CartaX, you can actually trade private stock without the mountain of paperwork usually required.

Key Limitations:

- **Users cannot completely depend on data:** It won't tell you if a startup is a unicorn or a bust; it just assumes you've already done the homework.
- **No Fact Checking:** Since it's operations-focused, it's not really looking out for fraud or double-checking if a founder's claims are legit.
- **Overkill for Newbies:** The platform is a bit of a beast and can feel way too complex if you're just starting out in the garage phase.

How It Helped Shape VentureIQ:

Carta proved to us that while solid back-office tools are essential, they only really handle the "aftermath" of

a deal. They miss the boat on the pre-deal insights that actually matter. That massive gap is why we built VentureIQ to bake evaluation tools right into the workflow, so you're getting the hard data and red flags *before* you sign anything.

Platform	Strengths	Weaknesses
AngelList	Investment execution, SPV management, fund pooling	No verification, minimal AI, limited quality analysis
Crunchbase	Rich data, advanced search and filtering by industry, location, funding status,, global coverage	No investment execution, Data Quality Issues, Free Tier Limitations
PitchBook	Deep market intelligence, deal terms, exit data	Institutional pricing only, no AI scoring, no execution
Gust	Early-stage focus, structured profiles, deal flow	No AI analysis, no cross-verification, no scoring
Carta	Cap table management, 409A, CartaX secondary trading	No investment intelligence, No fraud detection

	Investment execution	Data Verification	AI scoring	Fraud Detection	Investor matching
AngelList	YES	YES	NO	PARTIAL	NO
Crunchbase	NO	PARTIAL	YES	NO	NO
PitchBook	NO	YES	YES	NO	NO
Gust	YES	NO	NO	NO	PARTIAL
Carta	YES	NO	NO	NO	YES
VentureIQ	YES	YES	YES	YES	YES

Table 1: Comparative Analysis of Existing Platforms

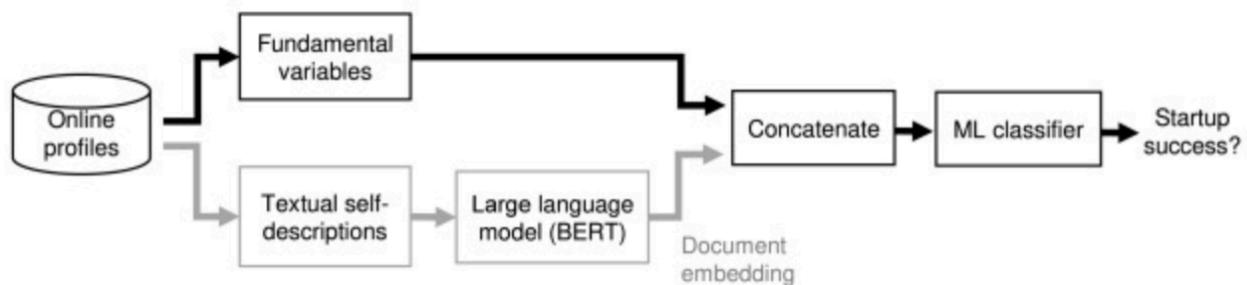
3.2 Related Research Papers

3.2.1 A Fused Large Language Model for Predicting Startup Success

Authors: Abdurahman Maarouf, Stefan Feuerriegel, Nicolas Pröllochs | European Journal of Operational Research, Vol. 322, Issue 1, April 2025 | DOI: [10.1016/j.ejor.2024.09.011](https://doi.org/10.1016/j.ejor.2024.09.011)

The researchers Maarouf Feuerriegel and Pröllochs 2025 trained their fused LLM model on 20,172 Crunchbase startup companies to forecast three business outcomes which included IPO, acquisition and external funding (seeding rounds). The study demonstrates that textual descriptions provide separate predictive value because market differentiation language and technology innovative language enhance prediction results which remain valid when revenue and funding history data are used as control variables. The fused multi-modal model consistently outperforms single-modality baselines across all three outcome categories.

This paper proposes a fused machine learning approach combining structured startup data and textual descriptions using a large language model (BERT) and traditional classifiers. The study demonstrates that integrating textual data significantly improves prediction accuracy from 72% to 74.33% and investment returns, achieving an AUROC (Area Under the Receiver Operating Characteristic Curve i.e. a performance metric used to evaluate binary classification models by measuring their ability to distinguish between positive and negative classes.) of 82.78% and a 7.23x ROI (Return of Investment).



Classifier	Balanced accuracy	Precision	Recall	F_1 -score	AUROC	AUCPR	ROI
Logistic Regression	73.71 (0.40)	58.51 (0.47)	78.79 (0.48)	67.15 (0.43)	81.76 (0.31)	72.18 (0.59)	704.98 (6.43)
Elastic net	74.27 (0.21)	58.61 (0.31)	80.43 (0.22)	67.81 (0.22)	82.51 (0.25)	73.11 (0.50)	706.30 (4.30)
Random forest	73.69 (0.64)	58.28 (0.90)	79.28 (1.68)	67.16 (0.70)	81.75 (0.63)	72.17 (0.82)	701.72 (12.37)
Our fused large language model (neural network)	74.33 (0.25)	59.83 (1.79)	78.28 (2.63)	67.77 (0.15)	82.78 (0.25)	73.70 (0.49)	723.09 (24.56)

Key Features:

- **Hybrid Data Fusion (Text + Structured Data):** Combines numerical metrics with startup descriptions using LLM embeddings, improving prediction accuracy beyond traditional models.
- **Use of LLM (BERT) for Semantic Understanding:** Extracts deeper meaning from startup descriptions instead of relying on keywords, capturing vision, positioning, and business clarity.
- **Explainability with SHAP:** Provides interpretable insights into why a startup is predicted to succeed or fail, increasing trust for decision-making.
- **Large-Scale Real Dataset (Crunchbase):** Uses 20K+ startups, making results more realistic and applicable to real-world startup ecosystems.

Key Limitations:

- **No Data Verification Layer:** Relies on self-reported Crunchbase data, which may be inaccurate.
- **No Investor–Startup Matching Intelligence:** Lacks behavioral or graph-based matching, which is critical for connecting the right investors with startups.
- **Static Data (No Real-Time Updates):** Uses historical data only, whereas real investment decisions require continuously updated metrics
- Large language models such as BERT may embed biases that are populated in downstream tasks. As these models are trained on vast corpora of text data, which inevitably contain societal biases.

How It Helped Shape VentureIQ:

The study demonstrated that researchers should use both organized and unorganized startup information together with financial data. The research confirmed our hypothesis about implementing document intelligence and AI-driven startup analysis within VentureIQ. The research results shaped our method for multi-dimensional scoring and contextual startup evaluation through AI model evaluation.

3.2.2 Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG

Authors: Aditi Singh, Abul Ehtesham, Saket Kumar, Tala Talaei Khoei, Athanasios V. Vasilakos | Source: [arXiv:2501.09136](https://arxiv.org/abs/2501.09136) | Published: January 2025

The paper introduces Agentic RAG systems through their definition which describes these systems as AI models that generate responses while they determine their methods for information retrieval tool selection and stepwise reasoning execution. The paper studies different architectures of Agentic AI systems and their impact on the reliability and contextual quality of AI-generated outputs.

Key Features:

- Explains how Agentic AI systems make autonomous decisions.
- Combines Retrieval-Augmented Generation (RAG) with external tools and reasoning.
- Improves contextual understanding using dynamic information retrieval.
- Provides different architectures and workflows for Agentic RAG systems.

Key Limitations:

- Agentic systems can become computationally expensive.
- Performance depends heavily on retrieval quality and external data sources.
- Complex architectures increase implementation difficulty.
- Risk of generating incorrect outputs if retrieval is unreliable.

How It Helped Shape VentureIQ:

The research paper established fundamental design principles which built the core system architecture used in VentureIQ. The system enabled our team to develop an Agentic AI component which automatically collects startup data from various sources without requiring input from users. The paper provided evidence that supported our decision to implement RAG as the method for producing contextual startup insights through historical data and market information.

3.2.3 FinBERT: Financial Sentiment Analysis with Pre-trained Language Models:

Author: Dogu Araci | arXiv | August 2019 | DOI: Not officially assigned (arXiv preprint) | arXiv ID: 1908.10063 | <https://arxiv.org/abs/2006.08097>

This paper pre-trains a finance-specific BERT model using 4.9 billion tokens that come from corporate 10-K and 10-Q filings, plus earnings call transcripts and also analyst reports, basically the material general-purpose BERT was never really exposed to. In their work they test four FinBERT variants across three financial sentiment classification datasets, namely Financial PhraseBank, AnalystTone, and FiQA. The top result, the uncased FinBERT-FinVocab variant, ends up improving over standard BERT by as much as 29.2% on FiQA .

Key Features:

- Pretrained on 4.9B tokens of real financial text
- Four variants: cased/uncased × BaseVocab/FinVocab
- Beats BERT by up to 29.2% on FiQA
- Analyst reports outperform larger corpora proportionally

Key Limitations:

- Trained only on US SEC filings and English text
- Startup pitch language is largely absent from training data
- Sentiment classification only - not built for claim extraction
- FinVocab advantage is marginal on most benchmarks

How It Helped Shape VentureIQ:

Domain-specific pretraining clearly outperforms general BERT on financial text, which pushed us toward fine-tuning on a curated Indian startup corpus rather than using an off-the-shelf model for VentureIQ's document intelligence layer.

	BERT		FinBERT-BaseVocab		FinBERT-FinVocab	
	cased	uncased	cased	uncased	cased	uncased
PhraseBank	0.755	0.835	0.856	0.870	0.864	0.872
FiQA	0.653	0.730	0.767	0.796	0.814	0.844
AnalystTone	0.840	0.850	0.872	0.880	0.876	0.887

Table 2: Performance of different BERT models on three financial sentiment analysis tasks.

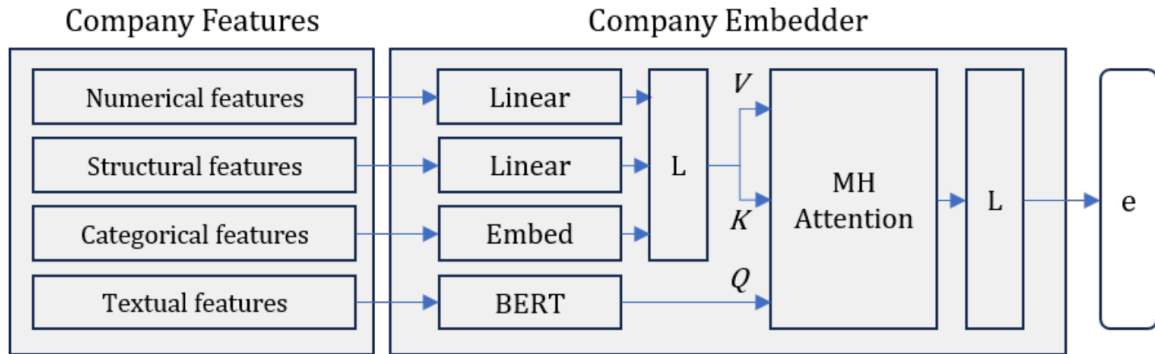
		10-Ks/10-Qs	Earnings Call	Analyst Reports	All
BaseVocab	PhraseBank	0.835	0.843	0.845	0.856
	FiQA	0.707	0.731	0.744	0.767
	AnalystTone	0.845	0.862	0.871	0.872
FinVocab	PhraseBank	0.847	0.860	0.861	0.864
	FiQA	0.766	0.778	0.796	0.814
	AnalystTone	0.858	0.870	0.872	0.876

Table 3: Performance of pretraining on different financial corpus.

3.2.4 Predicting Startup-VC Fund Matches with Structural Embeddings and Temporal Investment Data

Source: [arXiv:2511.23364](https://arxiv.org/abs/2511.23364) | Published: November 2024 | Dataset: Uzabase SPEEDA (~24,000 startups, ~100,000 investors)

The research predicts whether a specific VC fund will invest in a specific startup — treating it as a direct yes/no decision for each fund-startup pair rather than a general ranking system. Each startup is represented using its text description, basic numbers, industry tags, and network position within the investment ecosystem. Each fund is represented by its last 15 investments in time order, capturing how its preferences have evolved. Validated on Japanese startup data from the SPEEDA platform covering 24,000 startups and 100,000 investors.



Key Features

- Uses Node2Vec to capture each startup's position in the investment network who co-invested, how connected it is
- Represents each VC fund as a time-ordered sequence of past investments using LSTM to track how fund preferences change over time
- Combines four information types text, numbers, industry tags, network position using Multi-Head Attention to weigh what matters most per startup
- Achieved F1 score of 0.75, a 7 percentage point improvement over the baseline using basic startup attributes alone

Key Limitations

- Brand new startups with no investment history cannot be properly represented in the network
- Node2Vec embeddings are static they do not update automatically as new investments happen
- Model depends heavily on past behaviour sudden strategy changes in a fund may not be captured quickly
- Currently tested only on Japanese data performance in other markets is unverified
- Limited interpretability: hard to explain clearly why a specific pair was predicted as a match

How It Helped Shape VentureIQ

- The core insight — that a startup's network connections matter as much as its own profile — directly inspired our Neo4j-based graph analysis of co-investment relationships
- The fund's historical investment sequence approach shaped how VentureIQ models investor behaviour over time
- Unlike this paper, VentureIQ extends further by adding real-time signals and startup verification areas the paper itself flags as future work

4. Problem Definition

The ecosystem of startup investment has seen significant developments in recent years, yet the methods for evaluating start-ups are based on different sources of data and independent research and assessments. Investors gather information through various channels such as Crunchbase and AngelList, as well as LinkedIn, websites and pitches of startups. Yet, there is no reliable mechanism for the verification of the authenticity of startup data, which differs from source to source. Thus, the investment decision is based on incomplete data, which is provided by organizations themselves, leading to increased risks of investing in unsuccessful initiatives, whereas potentially successful ones may go unnoticed due to lack of visibility.

Based on the analysis of existing platforms and scholarly studies, it can be seen that current systems focus on a particular aspect of the startup investment process. Such services as Crunchbase and Pitchbook provide information regarding start-up and the market, respectively. AngelList and Gust, in turn, pay more attention to the fundraising process and relationships with investors. Finally, after investment decisions are made, Carta provides equity management solutions. Currently, there is no unified platform integrating information collection, artificial intelligence startup evaluation and detection of frauds, as well as intelligent recommendations for investment decisions.

Similarly, other research papers analysed during the course of this project have underlined the rising significance of Agentic AI, Retrieval-Augmented Generation (RAG), financial NLP models, and graph recommendation systems in enhancing startup evaluation and investor decision-making.

The research problem that VentureIQ seeks to solve is:

"How could a startup investment intelligence platform that integrates Agentic AI, Retrieval-Augmented Generation (RAG), and machine learning be designed to offer accurate evaluation of startups, fraud detection, intelligent matching of investors to startups, and investment support?"

The **key sub-problems** addressed by VentureIQ include:

- Reliance on startups' self-reported information with no independent verification.
- No standardized process for evaluating startups' performance and capabilities.
- Difficulties in detecting fraudulent activities, discrepancies, or misleading information.
- High level of information asymmetry between entrepreneurs and investors.
- Efficiency issues with finding startups and matching them with investors.
- No decision support systems built into current platforms based on AI.
- Absence of a unified platform with features of verification, analytics, and investment assistance.

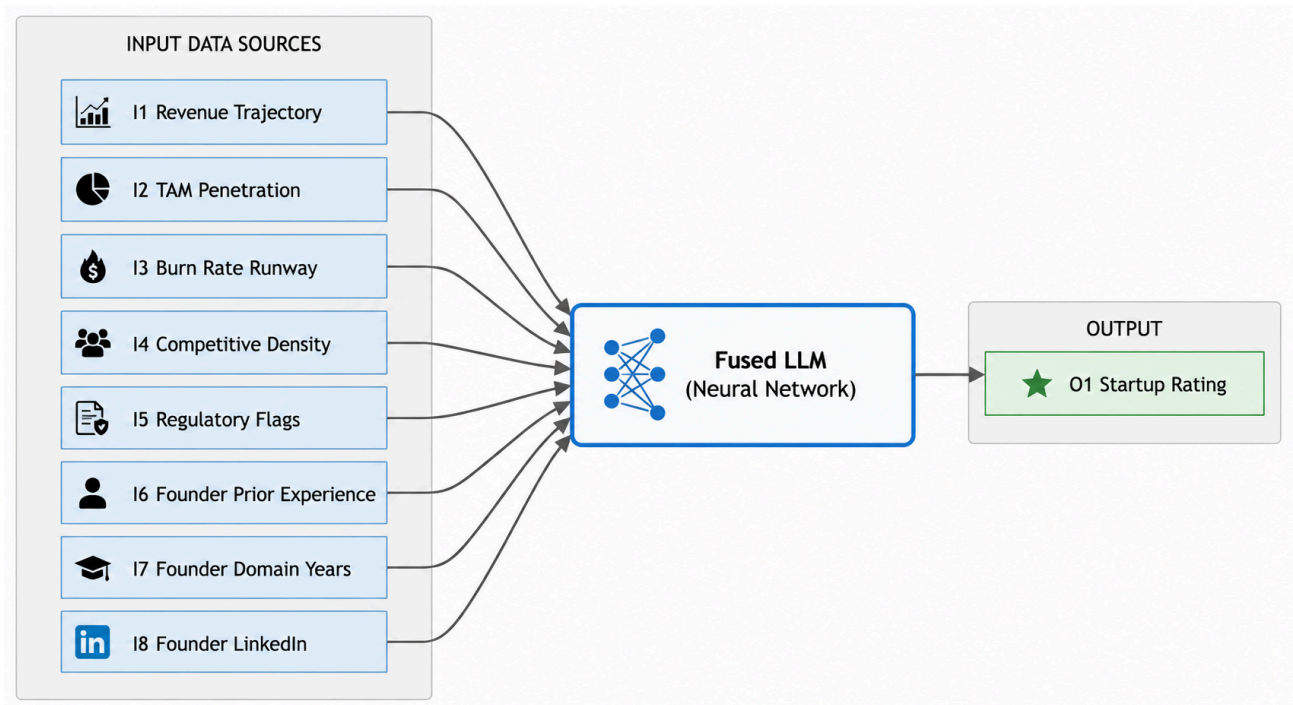
VentureIQ will try to tackle these problems by integrating multi-source information verification, AI-driven analytics, anomaly detection, behavioral recommendation systems, and transparent evaluation of startups."

Therefore, the core problem is to design a system that enables reliable, transparent, and data-driven evaluation of startups while facilitating efficient and secure investment interactions between startups and investors, reducing uncertainty, improving trust, and supporting better decision-making through the integration of verified data pipelines, intelligent analytics, and structured investment mechanisms.

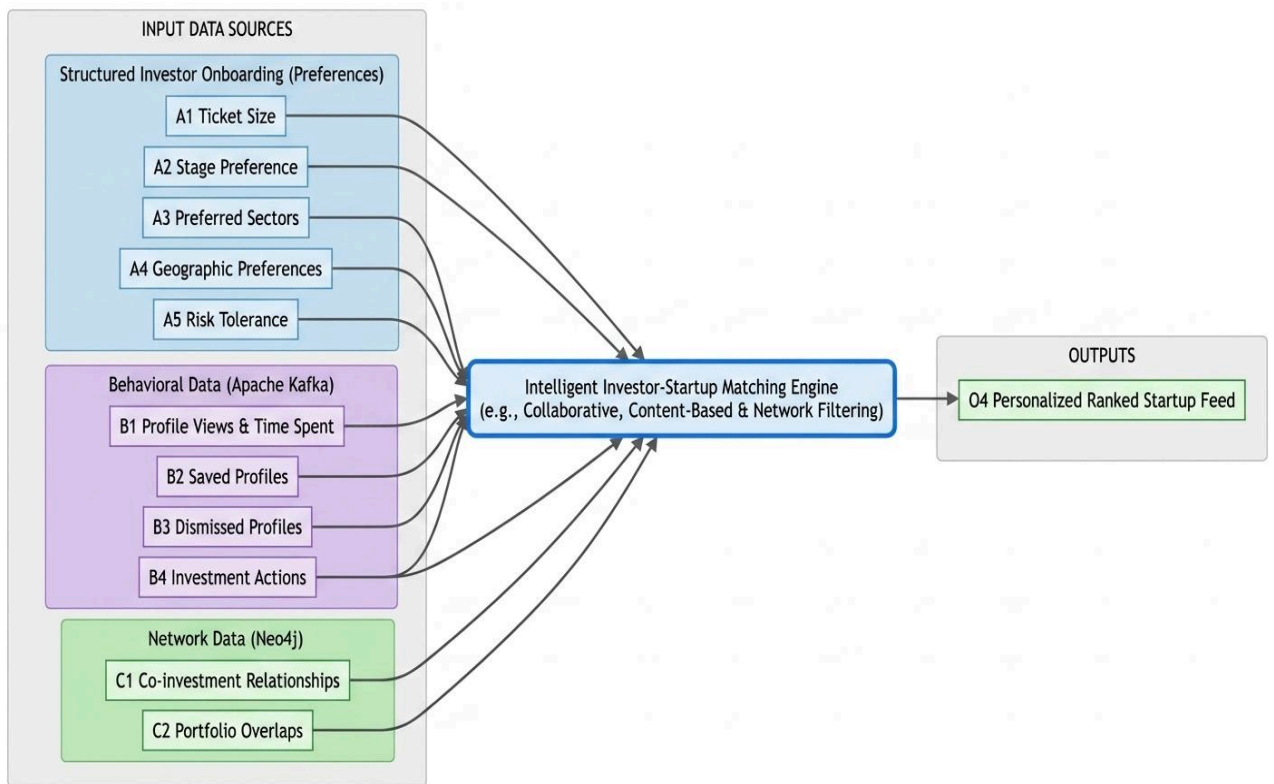
5. Objectives of the Project Work

VentureIQ is designed around the following core objectives:

1. **Develop an AI-Based Multi-Dimensional Scoring System:** Design models to evaluate startups across multiple dimensions like growth potential, risk level, market opportunity, etc.



2. **Reduce Information Asymmetry:** Enable data transparency and comparability across startups and ensure investors & startups operate with equal access to information.
3. **Implement Advanced Risk & Fraud Detection Mechanisms:** Use anomaly detection models to identify Inconsistent financial data, Sudden metric spikes and Suspicious behavioral patterns.
4. **Enable Intelligent Investor–Startup Matching:** Build recommendation systems using Investor preferences and Behavioral data.



6. Requirement Specification

6.1 Functional Requirements

6.1.1 Startup Data Pipeline

- The registration forms are designed with stage-awareness. An idea-stage startup does not have any place to fill out its burn rate. Such conditional fields are made possible through Zod validation.
- Metrics collected as financials via individual profile: Revenue, burn rate, runway, CAC (customer acquisition cost), LTV (Lifetime Value), TAM (total addressable market), and SAM (serviceable addressable market). We chose these metrics specifically because it would be needed for an investor to calculate basic unit economics.
- Pitch decks, cap tables, and other financial statements are parsed by OCR (optical character recognition) technology. LayoutLM was specifically chosen because it can understand document layouts in addition to raw text content.
- An agentic enrichment component cross-checks the self-reporting done by the founder on other platforms like LinkedIn, MCA21, GSTN and GitHub. That's because self-reported information is not good enough for investors.
- A vector store built on Pinecone indices embeddings of past startups and industry benchmarks. This allows our RAG system to use the context from other similar startups when evaluating a new one.
- An Isolation Forest and TensorFlow Autoencoder are used to screen each profile. Profiles which cannot be reconstructed back to within a certain error threshold are flagged for manual inspection.

6.1.2 VC/Investor Data Pipeline

- Ticket size, stage preference, preferred sectors, geographical preferences, and risk tolerance for investors are all obtained from investor onboarding. The inputs provide the initial recommendations based on structured data until behavioural data is collected.
- All user interactions i.e. profile view, time spent on profiles, saved profiles, dismissed profiles—are sent through an Apache Kafka stream in real-time. Kafka has been used over a database since Kafka processes all events concurrently without blocking the frontend.
- The feed generation uses a Collaborative Filtering algorithm based on LightFM using both structured and behavioural features to build personalized feeds.
- A graph built using Neo4j connects investor profiles and startups based on interactions, showing sectors that an investor prefers along with investment co-relations within a network.

6.1.3 AI Output Layer

- The scoring engine produces four independent scores which include Growth Potential and Risk Level and Fraud Likelihood and Founder Credibility. The Investor Dashboard combines different elements which include scores and match recommendations and real-time fraud alerts and portfolio tracking.
- The system requires a load time of less than two seconds while it needs a usability score of at least 70 according to UAT participants (User Acceptance Testing - these are real users who test the system before launch.).

6.2 Non-Functional Requirements

- **Scalability:** The platform shall scale horizontally to handle growing numbers of startups, investors, and data processing workloads without performance degradation.
- **Reliability:** The system must achieve high availability supported by robust logging, monitoring, and observability infrastructure.
- **Security:** All user data, financial information, and documents shall be encrypted at rest and in transit; KYC and escrow mechanisms shall ensure regulatory compliance.
- **Explainability:** All AI-generated scores and recommendations shall be accompanied by human readable SHAP rationale to support investor trust and regulatory accountability.
- **Usability:** The interface shall be intuitive for both first-time angel investors and experienced venture capital fund managers.
- **Performance:** Scoring and recommendation queries shall respond within acceptable latency thresholds to ensure a fluid investor experience.

6.3 Technical Stack

Component	Technology / Tool
Frontend	React with schema-driven dynamic forms (stage-aware conditional fields)
API Layer	RESTful API Gateway with Auth and KYC middleware
Structured Storage	PostgreSQL for normalised startup and investor profile data
Document Storage	S3-compatible Blob Storage for pitch decks, cap tables, and financials
Vector Database	Pinecone / Weaviate for RAG knowledge base — benchmarking and pattern detection
Graph Database	Neo4j for Investor–Startup Network Graph and co-investment analysis
Event Streaming	Apache Kafka pipeline for real-time investor behavioral event tracking
ML / AI	Scikit-learn and PyTorch for model training; SHAP for explainability; LayoutLM for document parsing
Fraud Detection	Isolation Forest and Autoencoder-based unsupervised anomaly detection

Table 2: Technical Stack Summary

7. System Design

VentureIQ is organised into five layers. Each layer has one job, and they're sequenced deliberately such that no layer does anything until the one before it has done its job cleanly.

Layer 1 : Infrastructure

All data coming in and out of the platform goes through the API Gateway – each request, no matter what user category, always passes through it. There is no way around it.

Then comes Auth & KYC that is immediately behind the Gateway. Auth deals with authentication through use of JWT tokens. KYC is concerned with verification of legitimacy of both the customers' identities and their businesses – without that you won't get away with building a financial platform.

The Notifications component will handle delivery of emails and alerts for all platform processes: suspicious user profiles, investments made, notifications in your dashboard.

Monitoring (using Prometheus and ELK stack) will track all latencies, error rates, and traffic going through the system.

Layer 2 : Startup Data Pipeline

The startup data enters this location. The system receives unstructured data inputs which create structured data outputs after going through validation.

Input Forms & Document Uploads - The startup's founders fill out the structured input forms (revenues, growth, team, market size) while they submit pitch decks and capitalization tables in PDF format.

Validation & Parsing - The system identifies all input data as invalid until further verification occurs. The validation module verifies all fields and possible errors. OCR and NLP technologies extract all relevant information from the PDF files.

Agentic Enrichment - The autonomous system collects information from LinkedIn, GitHub, company records, and financial databases to validate founder claims and complete information that the founder failed to provide.

Data Storage - All structured information is stored in PostgreSQL; all raw input is stored on AWS S3.

Layer 3 : AI Engine

Investor intelligence leverages startup data that is translated into investment intelligence by the following system.

Feature Store transforms database tables into feature vectors through normalised number conversion and category encoding and calculation of financial ratios. Pre-processing is done here since models need clean data rather than unstructured data.

The system takes features of four scoring models to calculate the final score.

Growth Prediction – forecasts revenue trends based on historical data

Risk Assessment – determines burn rate, runway, and overall company health

Founder Credibility – Prior exits, domain years, LinkedIn, Github commits etc.

Fraud Detection – detects inconsistencies between provided data and enriched data
The AI engine outputs a final score which reflects the financial viability / chances of success for the startup.
This score is used by investors to make investment decisions about specific startups.

Layer 4 : Investor Pipeline

The startup pipeline works parallel to the investor pipeline.

In the setting up of investor profiles, the system prompts the user to indicate their preferences in sectors and the stage of investment, along with the normal ticket sizes. The structured input functionality helps make cold start recommendations by the system even before any data regarding the behavioral input is generated.

The system keeps track of the behavior of the user by tracking clicks, tracking time spent on profile and investments, and then feeding it to the event logs using Apache Kafka in real-time.

The recommendation engine makes use of LightFM (Python library for building hybrid recommendation systems that combines collaborative filtering and content-based filtering techniques) collaborative filtering to personalize startup feeds depending on the behavior and preferences of the user, while using the same content recommendation techniques to give deal recommendations.

Investor Dashboard acts as an interface for VC and angel investors, displaying startups on the basis of AI scores, giving detailed info about scores and metrics and uploading documents. All processes for showing interest on the platform can be accomplished by the user in one go from this dashboard.

Layer 5 : Marketplace and Compliance

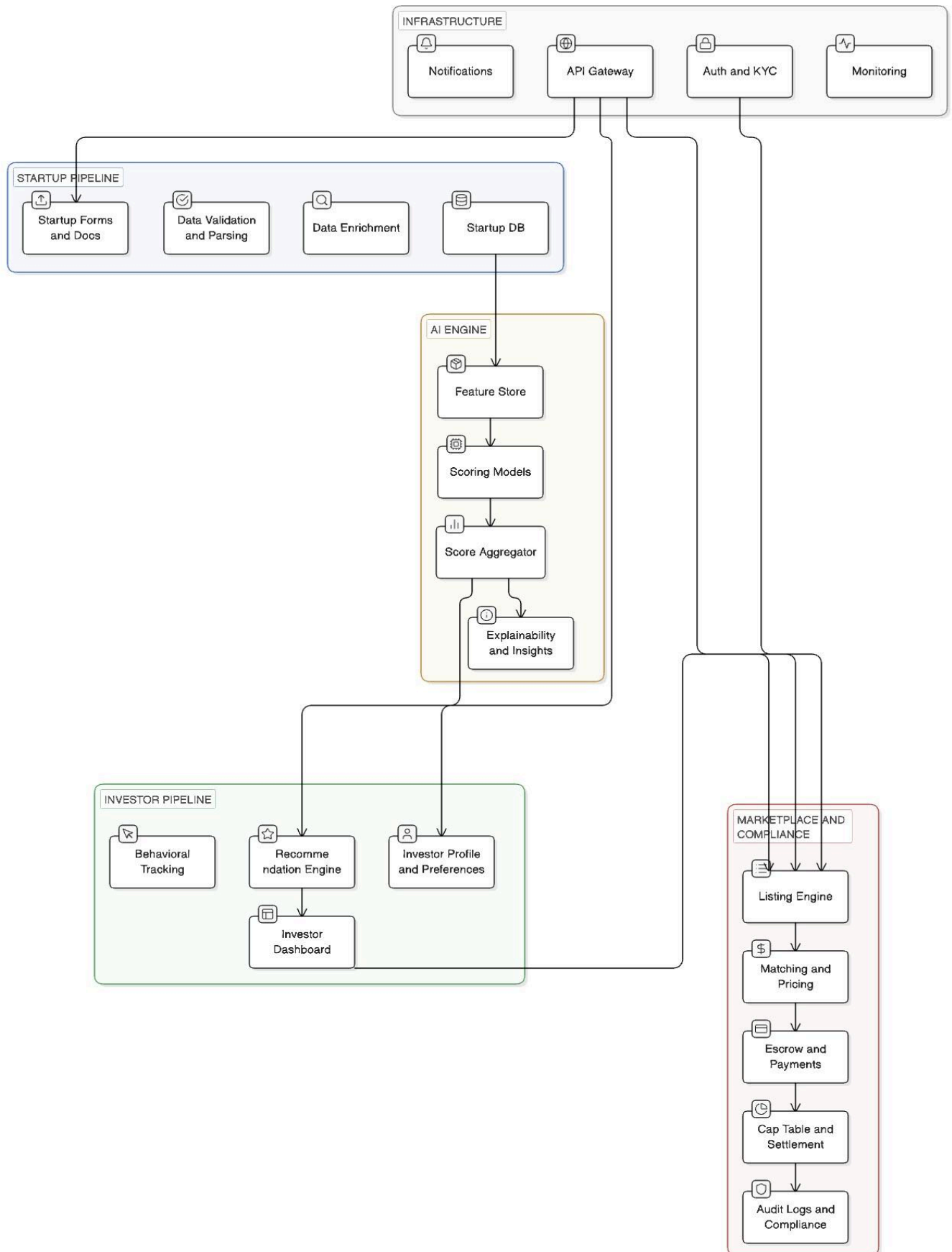
This is where interest becomes a transaction.

The main fundraising process involves the usage of a listing engine where startups apply for funding. This leads to an official marketplace listing being created.

The output of the recommendation engine becomes the input for the matching between investors and listings; meanwhile, Fair Value Estimator works at the same time, providing per-share price estimation through AI scoring and comparing transactions.

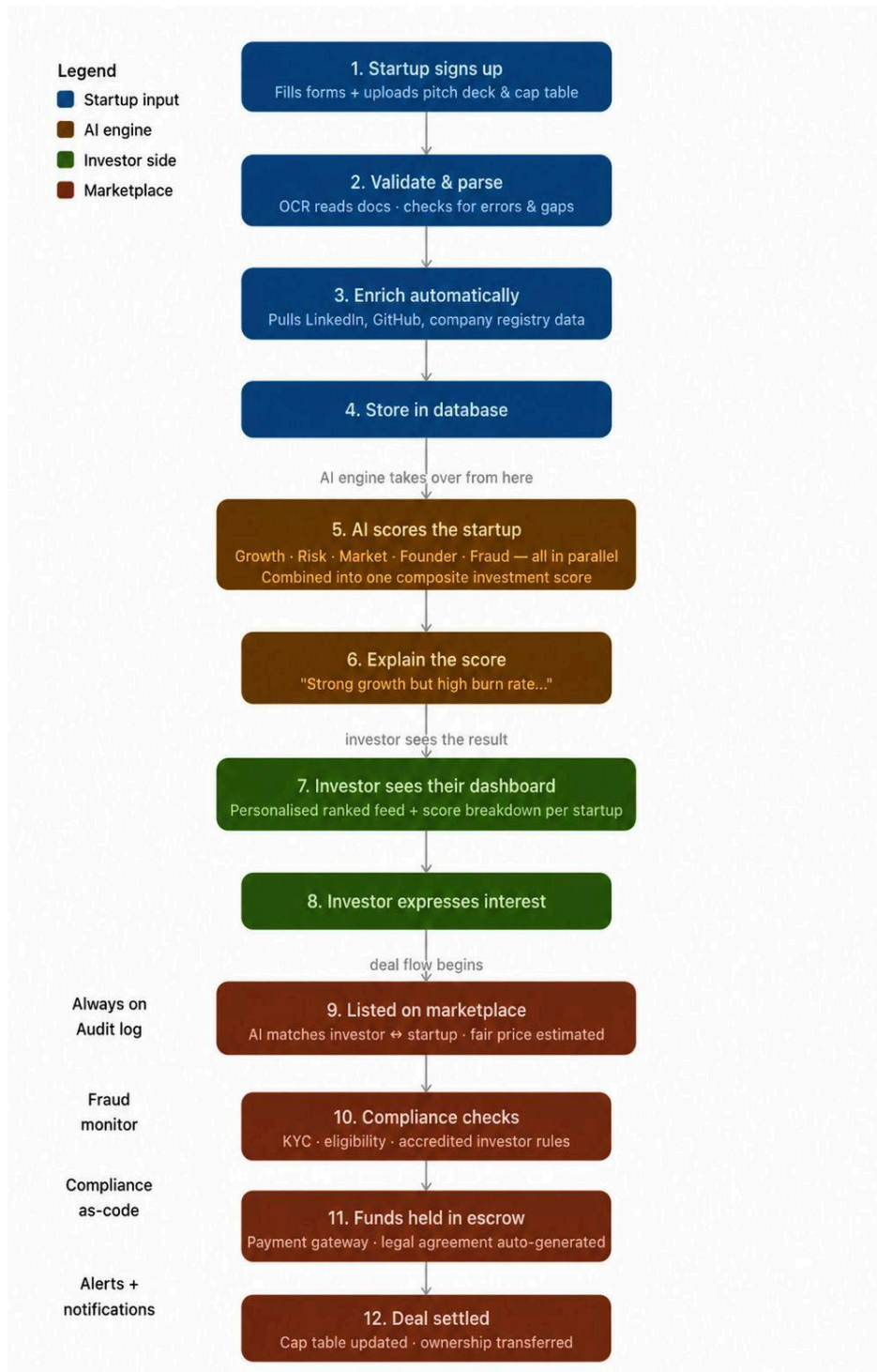
Funds collected in the escrow system account until all compliance checks are conducted since direct money transfer from one party to another does not take place. Payment gateway proceeds with transferring money only after completion of all the necessary checks. Ownership records will be updated in real-time, reflecting the new ownership stake percentage for the respective investor.

A fixed audit log containing all actions during the whole course of the transaction process is created by the system. The Compliance-as-Code layer carries out automatic check of rules in order to meet SEBI regulations and accredited investor rules and ownership limitations.



8. Methodologies

VentureIQ system design includes two merging data pathways: the Startup Data Pipeline and the VC/Investor Data Pipeline. The system connects both pathways using common services and an artificial intelligence output system. These two pathways operate a central Feature Store, which supplies data to the AI Scoring System and delivers results through the Investor Dashboard.



8.1 Startup Data Pipeline

8.1.1 Structured Input — Manual and Guided Forms

- Forms are stage-conditional i.e. Idea, Seed, and Growth stages each render different fields using React with Zod validation (TypeScript-first schema validation library). Asking a pre-revenue founder for burn rate payback data is pointless, so we don't.
- Backend re-validates every submission through Pydantic v2 before anything touches the database - two validation layers, client and server.
- Metrics collected: revenue, burn rate, runway, TAM, SAM, CAC, LTV, active users, plus document uploads for pitch decks, cap tables, audited financials, startup age.

8.1.2 Document Intelligence Layer

- Tesseract or Paddle OCR extracts raw texts, while LayoutLM-v3 (uses a BERT-based architecture enhanced with 2D positional embeddings to capture word coordinates and visual embeddings to recognize font styles, boldness, and other visual cues) further analyses it using spatial cognition
- This is important since financial documents use tables to structure their information. An algorithm that only analyses text in a linear manner without spatial comprehension would read the data incorrectly since it is structured into columns
- The Spacy NER (named entity recognition) module, which recognizes and classifies entities such as company names, dates, and numbers, operates after extraction

If the extracted number deviates by more than 20% from what was submitted by the founder in the form, it raises a red flag before scoring

8.1.3 Agentic AI Enrichment

By agentic in this instance we mean that the algorithm is responsible for deciding what data sources need to be queried, and the sequence in which they need to be queried, unlike an API integration where there is a pre-defined script that gets executed.

- **LinkedIn:** Employment history, previous businesses, expertise in the field of founders.
- **MCA21:** Verification of the incorporation and registered address as per Ministry of Corporate Affairs database.
- **GSTN:** Cross verification of claimed revenue through the tax payment history.
- **GitHub:** Number of commits as well as number of contributors in case of a technical team.
- All calls will have exponential back off along with five retries, and will fallback to cached results from the last thirty days.

8.1.4 RAG-Based Benchmarking System

RAG means “Retrieval-Augmented Generation,” and in this context, it implies that rather than rating the startup according to the knowledge of the AI, the system retrieves reference material to make use of it for rating the startup.

The Pinecone Vector Database (data stored mathematically, rather than keyword-based search) contains the

profiles of all past startups, including industry benchmark reports. The startup is rated by making comparisons with the most similar ten out of all historical profiles, and not all. Thus, a Seed fintech startup is compared with other Seed fintech startups, and not Series B SaaS startups. Anything older than 24 months is deprecated.

8.1.5 Fraud Detection Layer

- **Isolation Forest:** unsupervised approach to outlier detection based on the efficiency of its isolation process in random decision trees. Outliers are identified faster because of their inherent structural difference when compared to most of the other data.
- **TensorFlow Autoencoder:** trained on legitimate data points, it can learn the patterns of legitimate data. Reconstruction error will be high for the new data point, indicating that it is not a part of the pattern it learnt during training – the fraud.
- Areas targeted for fraud detection include: sudden increase in income levels without any changes in GST rate, financial ratios that do not fit within the industry norms, discrepancies between the information provided in forms, documents, and API responses
- Algorithms identify frauds; humans perform further verification and take final actions on new profile approval

8.2 VC/Investor Data Pipeline

8.2.1 Structured Investor Profiling

- Represents ticket size, event preference, industry preference, geography preference, and risk tolerance at onboarding
- The profile defined here forms the foundation of the cold start problem — if we don't have it, our recommendation engine has no way of knowing what new users want

8.2.2 Behavioral Event Tracking

- Apache Kafka (event streaming platform - handles large data volumes of real-time traffic between systems without congestion) records all the actions taken by the investor: viewing the profile, time spent on each section, viewing, saving, dismissing documents, making investments
- Database logging directly would create concurrency problems if there were several investors using the system at once; Apache Kafka neatly separates events creation from consumption
- The events are then loaded into the event_log table in the PostgreSQL database to train the recommendation engine

8.2.3 Preference Learning and Recommendation Engine

- The algorithm LightFM operates on Collaborative Filtering. It finds those investors who demonstrate similar behavior patterns and assumes that they will exhibit similar behavior in case they are confronted with the same startups.
- There is no batch processing, but continuous updates. In the case where an investor turns down three successive fintech startups, he/she should not be shown a fourth one.
- The last feed is ranked using Neo4j network score

8.2.4 Investor–Startup Network Graph

A Neo4j graph database builds and maintains an Investor–Startup Network Graph mapping co-investment relationships, portfolio overlaps, and investment sequencing patterns, enabling the platform to:

- Identify high-conviction investors in specific sectors
- Surface co-investment opportunities between complementary investors
- Understand network-level dynamics that historically predict funding success



Figure 4: Investor–Startup Matching Pipeline — Five-Stage Flow

8.3 AI Output and Scoring System

Both data pipelines converge on a unified Feature Store that consolidates all engineered features. The AI Scoring System applies weighted aggregation across four primary scoring dimensions:

- **Growth Potential Score:** Revenue growth trajectory, user growth trajectory, total addressable market percentage captured, and FinBERT text indicator based on the founders’ market opportunity claim
- **Risk Level Score:** Burn rate runway in weeks, competitive landscape intensity computed using RAG-retrieved benchmarking data, and regulatory risk flag per industry
- **Fraud Detection Score:** The combination of Isolation Forest score and Autoencoder score computed in the pipeline detailed above
- **Founder Credibility Score:** The number of exits, years of experience in the domain, number of LinkedIn endorsements, and GitHub commits per week
- Four such scores are computed independently of one another due to the inherent statistical characteristics of qualitative and financial data and suboptimality that results from combining them too early, as demonstrated in the ablation studies by Maarouf et al. (2025)
- **SHAP:** SHAPley Additive explanations, a method rooted in game theory. It computes the three most important indicators contributing to each score separately and then converts them into text rationales

in English on the dashboard. In effect, it allows the investor looking at a poor Founder Credibility score to see why exactly their score is poor – either because of the lack of relevant expertise or of prior startup experience.

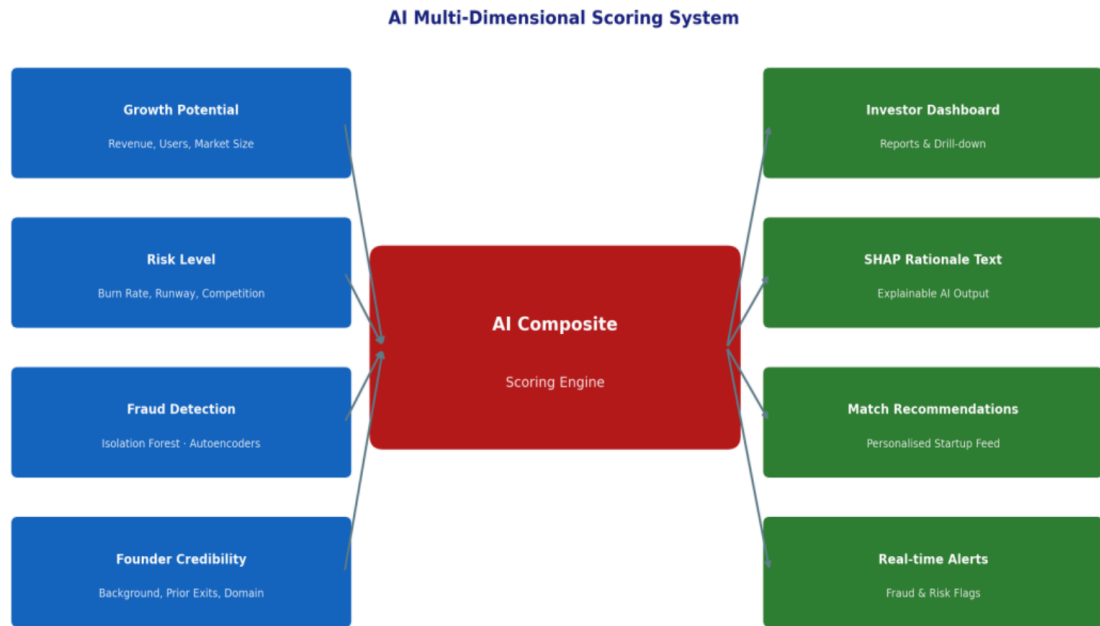


Figure 5: AI Multi-Dimensional Scoring System — Inputs, Scoring Engine, and Outputs

9. Conclusion

Investing in Indian startups faces a unique challenge with regard to the lack of the data infrastructure problem faced by no other platform. Crunchbase collects, AngelList acts, PitchBook analyzes, but none of them verifies. Venture capital investors on any of these platforms take a startup's founder at their word. And that is a fundamental problem, which grows exponentially in scale.

VentureIQ was built to fix that initially. Indeed, the whole report revolves around one main problem – how to make sure about a startup profile that one doesn't know personally? The answer turned out to require multiple actions, including verified data, document intelligence which verifies whether submitted documents are indeed verified against one another, enrichment through governmental APIs query independently, unsupervised machine learning algorithms to detect fraud without requiring the investor's help and SHAP-simulated scoring with explanations for each decision. Answer for the Explanation question was provided by SHAP paper, but not in such polite terms as one would use for a corporate presentation; there is a need, and not a matter of style – a single score could not make any sense for a person responsible for a decision regarding investment worth ₹50 lakh.

It should be understood that VentureIQ never intends to replace the human intuition of an investor. While the scoring engine will detect signals and raise the red flag, the decision will always remain human-made. There is no question about automated rejection, the fraud detection algorithm just escalates the case. SHAP explanation text explains without telling how to make the decision. In general, the system offers tools to organize the due diligence process in twenty hours, which is significantly faster during the first three hours.

The next step concerning the platform development involves filling the agentic system with data outside India, implementing a real-time news sentiment analysis as a dynamic indicator, and designing the marketplace from scratch into an asset transfer layer. All of that is described in the project v2.0 backlog.

10. References

- [1] A. Maarouf, S. Feuerriegel, and N. Pröllochs, A fused large language model for predicting startup success, European Journal of Operational Research, Vol 322, No 1, 2025, pp 198–214. [Link](#)
- [2] Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG, arXiv preprint arXiv:2501.09136, 2025. [Link](#)
- [3] D. Araci, FinBERT: Financial Sentiment Analysis with Pre-trained Language Models, arXiv preprint arXiv:1908.10063, Aug 2019. [Link](#)
- [4] Predicting Startup-VC Fund Matches with Structural Embeddings and Temporal Investment Data, arXiv preprint arXiv:2511.23364, 2024. [Link](#)
- [5] AngelList (Wellfound). <https://wellfound.com>
- [6] Crunchbase. <https://www.crunchbase.com>
- [7] PitchBook. <https://pitchbook.com>
- [8] Gust. <https://gust.com>
- [9] Carta. <https://carta.com>